342545

Course Name: Computer Networks (BCSC 0008)

Course Outcome

CO1: Understand the concept of OSI and TCP/IP reference model
CO2: Understand the basics of data transmission at physical layer.
CO3: Understand the channel allocation using ALOHA, CSMA and CSMA/CD.
CO4: Apply error detection and correction technique to eliminate transmission error.
CO5: Analyze the fixed and variable length address (IPv4) subneting for the given scenarios
CO6: Understand the design issues of the transport layer.
CO7: Understand the mechanism of protocols at application layer such as FTP, HTTP, Telnet, DNS.
CO8: Understand IPv6 addressing and differentiate it from IPv4.

Printed Pages: 3 **University Roll No. .....**

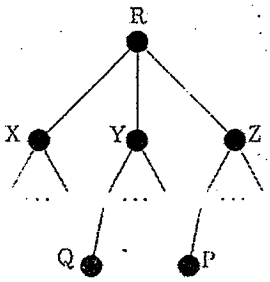
End Term Examination, Even Semester 2023-24
B.Tech (CSE, AIML, CCV, DA, CSF, IIOT), II Year, IV Semester
BCSC 0008: Computer Networks

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Consider a 100 Mbps link between an earth station (sender) and a satellite (receiver) at an altitude of 2100 km. The signal propagates at a speed of 10^8 m/s. Calculate the time taken (in milliseconds, rounded off to two decimal places) for the receiver to completely receive a packet of 1000 bytes transmitted by the sender.	4	3	A	C
2	Consider the cyclic redundancy check (CRC) based error detecting scheme having the generator polynomial X^3+X+1. Suppose the message $m_4m_3m_2m_1m_0=11000$ is to be transmitted. Check bits $c_2c_1c_0$ are appended at the end of the message by the transmitter using the above CRC scheme. The transmitted bit string is denoted by $m_4m_3m_2m_1m_0c_2c_1c_0$. Find the value of the checkbit sequence $c_2c_1c_0$.	4	4	A	P
3	Can you explain why the vulnerable time in ALOHA depends on transmission time but in CSMA depends on propagation time?	4	3	U	F
4	Compare the TCP header and the UDP header. List the field in the TCP header that are missing from UDP header. Give the reason for the absence.	4	6	U	F

5	Consider a computer network using the distance vector routing algorithm in its network layer. The partial topology of the network is shown below.  The objective is to find the shortest-cost path from the router R to routers P and Q. Assume that R does not initially know the shortest routes to P and Q. Assume that R has three neighbouring routers denoted as X, Y and Z. During one iteration, R measures its distance to its neighbours X, Y, and Z as 3, 2 and 5, respectively. Router R gets routing vectors from its neighbours that indicate that the distance to router P from routers X, Y and Z are 7, 6 and 5, respectively. The routing vector also indicates that the distance to router Q from routers X, Y and Z are 4, 6 and 8 respectively. What is the distance between R to P and R to Q, which will store in the table also write next hop router for a packet from R to P and R to Q.	4	5	E	P

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	A company has a class C network address of 204.204.204.0. It wishes to have three subnets, one with 100 hosts and two with 50 hosts each. What will be the network address and subnet mask of each network.	3	5	A	C

7	TCP is connection-oriented. It establishes a virtual path between the source and destination. Discuss the connection establishment process of TCP.	3	6	U	F
8	Write a short note on FTP, DNS and SMTP.	3	7	R	F
9	An IP datagram is carrying a TCP segment destined for address 130.14.16.17/16. The destination port address is corrupted and it arrive at destination 130.14.16.19/16. How does the receiving TCP react to this error?	3	5	A	P
10	What are the main differences between symmetric and asymmetric key cryptography, and how do these differences affect their respective applications in securing communication and data?	3	7	U	F

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Discuss the IP datagram format along with suitable diagram. Give a comparison of IPv4 and IPv6.	5	8	U	C
12	Classless Inter-domain Routing (CIDR) receives a packet with address 131.23.151.76. The router's routing table has the following entries: Prefix Output Interface Identifier 131.16.0.0/12 3 131.28.0.0/14 5 131.19.0.0/16 2 131.22.0.0/15 1 Identify output interface on which this packet will be forwarded.	5	5	A	P
13	An organization is granted the block 130.34.12.64/26. It needs to have 4 subnets. Answer the following 1) Number of IP address in each block 2) Network address and broadcast address of each block 3) First IP address of each block assigned to host 4) Subnet mask of each block.	5	5	An	P